THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL UNIT TEST SET 3

Chapters 1–4: **Exploring Substances: Acidic, Basic and Neutral, Electricity: Circuits and their Components, The World of Metals and Non-metals, Changes Around Us: Physical and Chemical** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. In the reaction above, the colour of the solution changes from: 1

(a) blue to colourless **(b)** green to blue
(c) blue to green **(d)** colourless to blue

2. A student heats a piece of wax and sees it melt. On stronger heating, black smoke appears. The black smoke indicates: 1

(a) no change **(b)** evaporation only
(c) melting only **(d)** unburnt carbon, a chemical change

3. A tablet that fizzes in water is likely to contain: 1

(a) a carbonate and an acid **(b)** only sugar
(c) only salt **(d)** only a base

4. A tester glows dimly with a pencil lead and brightly with a copper wire. This shows that: 1

(a) both conduct, copper better **(b)** only graphite conducts
(c) neither conducts **(d)** copper is an insulator

5. A student has four metals P, Q, R, S. P and Q react with dilute HCl, R and S do not. Q reacts faster than P. The most reactive is: 1

(a) P **(b)** Q **(c)** S **(d)** R

6. Why are bicycle chains regularly oiled? 1

(a) to make them brittle **(b)** to make them heavier
(c) to prevent rusting and reduce friction **(d)** to make them shiny only

7. A student adds drops of base to acid and notes the litmus. The sequence of colours is: 1

(a) red, purple, blue **(b)** blue, yellow, red
(c) red, green, blue **(d)** blue, purple, red

8. When a student removes one bulb from a string of decorative bulbs, all the bulbs go off. This shows that the circuit has: 1

(a) no cell **(b)** no wire
(c) a single path for the current **(d)** many separate paths

9. A metal M does not displace copper from CuSO4 but copper displaces it from M's salt solution. M is: 1

(a) as reactive as copper **(b)** less reactive than copper
(c) more reactive than copper **(d)** a non-metal

10. An experiment shows that the mass of a burnt candle's products (including gases) equals the mass of the candle and oxygen used. This supports: 1

(a) floating **(b)** rusting
(c) conservation of mass **(d)** melting

11. A student mixes equal volumes of acid and base. The mixture still turns blue litmus red. This suggests: 1

(a) the acid was stronger or more concentrated
(b) the mixture is neutral
(c) the base was stronger
(d) no reaction

12. If the bulb in the tester does not glow with iron nail, it could be because: 1

(a) nail is a plastic **(b)** iron is an insulator
(c) the nail is coated with paint or rust **(d)** cell is stronger

13. The mass of the iron nail after being kept in copper sulphate solution for long: 1

(a) becomes zero **(b)** decreases **(c)** increases **(d)** stays same

14. During the burning of a candle, heat and light are given out. This suggests: 1

(a) a chemical change **(b)** no change
(c) a physical change **(d)** a reversible change

15. A cook adds a pinch of baking soda to lemon-flavoured drink and it fizzes. This shows: 1

(a) salt reaction **(b)** no reaction
(c) acid-carbonate reaction **(d)** sugar reaction

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why do metals feel cold to the touch? 2

17. Why is a slow chemical change not always noticed? 2

18. Why does the stomach not get damaged by its own hydrochloric acid? 2

19. Why is the heating element usually a coil of nichrome and not of copper? 2

20. Explain why a gold ring does not need cleaning as often as a silver ring. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Describe a test to distinguish a piece of copper from a piece of aluminium without chemicals. 3

22. Calcium oxide reacts with water and the mixture becomes hot. Is it physical or chemical? Give two reasons. 3

23. Explain why a student's mixture of acid and base became warm. 3

24. Explain why a tungsten filament bulb glows white hot but an iron wire would not last. 3

OR

Why is a small piece of sodium kept in kerosene reduced in size when cut? 3

25. Explain with an example how a physical change may accompany a chemical change. 3

OR

Explain how red cabbage juice could be used to compare the strength of lemon and vinegar. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Explain, with the help of examples, how the ideas of acids, bases and neutralisation connect agriculture, health and the environment. 5

OR

Plan an energy-saving programme for a home with five measures and say why each helps. 5

27. Explain, with equations, the reactions of magnesium with oxygen, acid and copper sulphate solution. 5

OR

Analyse the changes taking place when you light a gas stove under a vessel of milk, with milk boiling over and burning at the bottom. Classify each change. 5

END OF QUESTION PAPER — Total: 50 Marks

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